DESCRIPTION OF COURSEWORK

Course Code	SWE 404
Course Name	Big Data Analytics
Lecturer	Toa Chean Khim
Academic Session	2024/04
Assessment Title	Assignment

A. Introduction/ Situation/ Background Information

This course provides a throughout introduction and current trend of the Big Data in industry field. The assignment comprises of 2 components. The first component involves writing a paper that explore the development trends of Big data in industry, as well as identify the other factors associated with its implementation. The second component involves presenting the finding from the paper.

B. Course Learning Outcomes (CLO) covered

At the end of this assessment, students are able to:

CLO 1 Describe the concept of Big Data.

CLO 5 Demonstrate the need to continually follow Big Data development trends.

C. University Policy on Academic Misconduct

1. Academic misconduct is a serious offense in Xiamen University Malaysia. It can be defined as any of the following:
   i. **Plagiarism** is submitting or presenting someone else's work, words, ideas, data or information as your own intentionally or unintentionally. This includes incorporating published and unpublished material, whether in manuscript, printed or electronic form into your work without acknowledging the source (the person and the work).
   ii. **Collusion** is two or more people collaborating on a piece of work (in part or whole) which is intended to be wholly individual and passed it off as own individual work.

iii. **Cheating** is an act of dishonesty or fraud in order to gain an unfair advantage in an assessment. This includes using or attempting to use, or assisting another to use materials that are prohibited or inappropriate, commissioning work from a third party, falsifying data, or breaching any examination rules.

2. All assessments submitted must be the student's own work, without any materials generated by AI tools, including direct copying and pasting of text or paraphrasing. Any form of academic misconduct, including using prohibited materials or inappropriate assistance, is a serious offense and will result in a zero mark for the entire assessment or part of it. If there is more than one guilty party, such as in case of collusion, all parties involved will receive the same penalty.

D. Instruction to Students

This assignment is a Group assignment (3 persons per group). Students must work together to create 1 report per group, but each individual will be evaluated separately later. The students should submit the following to the Moodle:

1) **Document A:** Assignment Cover Page, Report, and Marking Rubric in ONE report. Saved document as Group_No.docx and .pdf format. For example: Group_1.docx & Group_1.pdf.
2) **Document B:** Submit your Turnitin report by saved as Turnitin.pdf format.

Turnitin Class ID: [redacted]
Enrolment Key: [redacted]

3) **Google Form C:** Fill in the form for Individual Performance Evaluation using following links:

SWE Programme: https://forms.gle/vGjyibGaQSYY3CNP9

AIT Programme: https://forms.gle/o3M8sf8asSupLYih6

DMT Programme: https://forms.gle/isKiFEfzZbpfUUVc8

4) Group Leader needs to submit **Document A, Document B, and Google Form C** to the Moodle.
5) Other members in the group need to fill in **Google Form C**.

6) The submission deadline of Assignment is on **23:59, 31 May 2024.** Overdue penalty (5 marks deduction per day) will be given to the assignment that is submitted after the deadline. Submission type: Softcopy.

E. Evaluation Breakdown

No.	Component Title	Percentage (%)
1.	Abstract	10
2.	Introduction	10
3.	Main Content	40
4.	Discussion	15
5.	Conclusion	10
6.	Contribution	15
	TOTAL	**100**

F. Task(s)

The field of big data analytics has witnessed rapid growth and development in recent years, with significant implications for various industries and sectors. This assignment task aims to produce a high-quality report on the advancements and challenges in big data analytics. The group task is to conduct an in-depth review that critically examines the latest advancements, application, methodologies, and challenges in big data analytics. The following are some general guidelines. **It can be modified as per the format of your paper**.

i) Select a Topic:

Choose one specific emerging trend within the field of Big Data. Your chosen topic should be a recent development or innovation in Big Data. Ensure that there is enough academic literature available on your chosen topic.

ii) Abstract

- **Keep It Short**: Make your abstract brief, usually around 150-250 words.
- **Explain the Context**: Start by saying why big data is important and why you did the review. **(10)**

- **State Your Goal**: Clearly say what you aimed to do in your review.
- **Highlight Main Points**: Mention the key things you found in your review, like important trends or discoveries.
- **Say How You Reviewed**: If you used a special method, mention it briefly.
- **Explain Why It Matters**: Tell why your review is important for people who work with big data.

iii) Introduction

- **Start with a Hook**: Begin with an interesting statement or question to grab the reader's attention.
- **Explain Big Data**: Give a brief overview of what big data is and why it's important.
- **Highlight Relevance**: Discuss why studying big data matters and how it impacts various fields. **(10)**
- **State Your Review's Purpose**: Clearly say what your review paper aims to do.
- **Mention the Scope**: Describe what aspects of big data your review will cover.
- **Discuss Significance**: Explain why understanding big data is crucial.
- **Outline the Structure**: Briefly mention how your review paper is organized.

iv) Main Content *(Suggestion content for each part, encouraged to include additional relevant information)*

Big Data Technologies and Frameworks: Explore the technologies and frameworks commonly used in big data analytics, including Hadoop, Spark, and NoSQL databases.

Data Collection and Storage: Discuss methods and tools for collecting, storing, and managing large volumes of data, highlighting data quality considerations.

Data Processing and Analysis: Explain techniques and algorithms used for processing and analyzing big data, including batch and real-time processing. **(40)**

Machine Learning in Big Data: Explore how machine learning is integrated into big data analytics, with a focus on predictive modeling and data-driven insights.

Challenges in Big Data Analytics: Discuss common challenges and limitations in big data analytics, including data privacy, security, scalability, and ethical considerations.

Applications of Big Data: Provide real-world examples of big data applications in various domains, showcasing the impact on decision-making and innovation.

Future Trends and Emerging Technologies: Explore current trends and emerging technologies in big data research, such as edge computing, quantum computing, and AI-driven analytics.

v) Discussion *(Suggestion content for each part, encouraged to include additional relevant information)*:

Synthesize Findings: Synthesize the key insights from the main content, emphasizing the most significant contributions and trends in big data research.

Address Challenges: Discuss common challenges and limitations in big data analytics, such as data quality, scalability, and privacy.

Highlight Applications: Highlight real-world applications of big data in various domains and industries. **(15)**

Explore Emerging Trends: Discuss emerging trends, technologies, and methodologies in big data, such as AI integration and edge computing.

Discuss Ethical Considerations: Address ethical and legal considerations in big data, including data privacy, bias, and compliance with regulations.

Provide Recommendations: Offer recommendations for future research directions or practical applications based on your review.

vi) Conclusion

Summarize Key Points: Summarize the main findings from your review and their implications for the field of big data.

Reiterate Significance: Reiterate the significance of understanding and advancing big data research. **(10)**

Highlight Contributions: Highlight the contributions of your review paper and its value in synthesizing current knowledge.

End with a Forward-Looking Statement: Conclude with a forward-looking statement about the continued growth and impact of big data in the future.

vii) References

Cite Sources Properly: Ensure that you properly cite and reference all the sources used in your review paper following IEEE citation style. **AT LEAST 15 references.**

viii) Paper formatting:

The paper **AT LEAST 20 pages** in a single column format. Generate a similarity report in Turnitin that **not exceeding 20% similarity.**

ix) Individual Performance Evaluation:

Fill up the evaluation of each member in the group truthfully **(15)**

APPENDIX 1

MARKING RUBRICS

Component Title		**Assignment**					**Percentage (%)**	**20**
Criteria		**Score and Descriptors**					**Weight (%)**	**Marks**
		Excellent (5)	**Good (4)**	**Average (3)**	**Need Improvement (2)**	**Poor (1)**		
Group	**Abstract**	Engaging and informative abstract provided.	Clear and concise abstract provided.	Adequate abstract provided.	Vague abstract provided.	No clear abstract provided.	**10**	
	Introduction	Engaging and informative introduction provided.	Clear and concise introduction provided.	Adequate introduction provided.	Vague introduction provided.	No clear introduction provided.	**10**	
	Main Content	Engaging and informative main content provided, with comprehensive and insightful analysis and evaluation.	Clear and concise main content provided, with relevant analysis and evaluation.	Adequate main content provided, but lacking in-depth analysis and evaluation.	Vague or insufficient main content provided.	No main content provided or completely irrelevant.	**40**	
	Discussion	Engaging and informative discussion provided, with comprehensive and insightful analysis and evaluation.	Clear and concise discussion provided, with relevant analysis and evaluation.	Adequate discussion provided, but lacking in-depth analysis and evaluation.	Vague or insufficient discussion provided.	No discussion provided or completely irrelevant.	**15**	

	Conclusion	Engaging and informative conclusion provided, with comprehensive and insightful summarization of key findings and potential future directions.	Clear and concise conclusion provided, with relevant summarization of key findings.	Adequate conclusion provided, but lacking clear summarization of key findings.	Vague or insufficient conclusion provided.	No conclusion provided or completely irrelevant.	**10**				
Individual	**Contribution**	Consistently contribute to group. Effectively fulfill the assigned roles and responsibilities. The contribution enhances the overall outcome	Good contributed to group. Fulfill the assigned roles and responsibilities.	Adequate contributed to group. Adequate fulfill the assigned roles.	Limited contribution to group. Less adequate fulfill the assigned roles.	No contribution to group. Failed to fulfill the assigned roles.	**15**	**Student 1 Marks**	**Student 2 Marks**	**Student 3 Marks**	**Student 4 Marks**
						Total	**100**				

Note to students: Please include the marking rubric when submitting your coursework.